

Assessing Housing Affordability in Singapore with Bayesian Analysis

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SUMMARY

This report presents a Bayesian logistic regression model built to assess the affordability of houses in Singapore, particularly, the resale public housing market. The motivation behind the report stems from my own doubts and concerns of the rising prices of houses in Singapore. The model is built on a dataset that encompasses past transactions of resale public flats in Singapore from 2017 to 2024. The methods and theory covered are largely based on Bayesian Data Analysis by Gelman et al., 2021, supplemented with relevant papers regarding real estate analysis. A brief overview of the relevant statistical theories is included before diving into the setup of the model proper. The statistical theories are applied to real estate analysis by leveraging probabilistic programming libraries, like PyMC, which allow for sophisticated Bayesian methods to be easily utilised. Elaboration of some problems encountered on the way, that were resolved by suggestions from Bayesian Data Analysis, are included as well. The importance of model checks and evaluation were further emphasised after the building of the model. Concluding the report are the limitations and potential improvements for the model.

Although straightforward, this logistic regression model may potentially help with the big financial decisions that come along with buying a house. It may also be used as a stepping stone for more robust models in the future.

1. Introduction

With a staggering 89.7% home ownership rate in Singapore (Lim, A., 2024), buying a home comes as no surprise to many Singaporeans. However, home ownership has increasingly become a concern to young Singaporeans due to the unprecedented rising costs of living (Stolarchuk, J., 2023). For many young people, a house is likely the first big ticket item that they would have to buy, hence, careful planning is necessary to avoid financial distress. Understanding how much to save and which factors affect affordability of houses can provide some guidance in navigating the housing market. As a young Singaporean myself, I too have my doubts on being able to afford a house in the future with the seemingly exponential growth rate of house prices. Both Bayesian and non-Bayesian methods have been applied on real estate analysis and have yielded many meaningful results that may help an individual with the big decision of buying a home, with Bayesian models offering more flexibility and precision.

2. Background Information: Singapore's Housing Market

There are 3 main housing types in Singapore, mainly public housing regulated by the Housing Development Board of Singapore (HDB), condominiums and landed property which are both considered private properties.

HDB flats, or often abbreviated by just HDBs, are high-rise flats that come in various sizes, typically 1 to 5 bedroom flats, with generally homogenous layouts to ensure consistency across all units. There are over 1 million flats spread across the 24 towns in Singapore, with each HDB cluster consisting of an arbitrary number of blocks.



Figure 2.1(a): HDB cluster

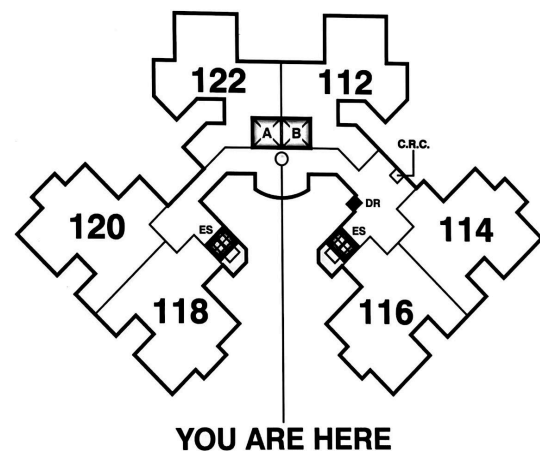


Figure 2.1(b): Floor Plan within Each Level of a HDB Block

HDB flats are known for their affordability and no-frills practicality, with a leasehold of only 99 years, after which is up, will be returned to the government. As they are regulated by HDB which is a government body of Singapore, there are certain restrictions when it comes to purchasing and selling HDB flats, e.g. there is an Ethnicity Integration Policy (EIP) and Singaporean Permanent Resident (SPR) quota for each block and neighbourhood to preserve racial harmony and for better assimilation of the PRs into local community respectively (HDB, n.d.). On the other hand, private properties are mostly freehold, and have little to no restrictions when it comes to buying and selling. Given that 77.8% of Singaporeans stay in HDB flats (Statistica, 2024), they will be the subject of this report.

Like all other real estate properties, there are many factors affecting HDB flat prices. Some macroeconomic factors include the overall economy of the country, like the inflation rate. Flat specific characteristics, like size and storey, also play a huge role in determining prices. The location of the flat is also crucial. Each town in Singapore also has its own unique characteristics, appealing to different demographics. However, since HDB flats are regulated by the government, there are some extra conditions that need to be laid out. As mentioned, HDB flats all come with a 99 year lease, thus, the amount of years left in the lease can highly impact the price.

Furthermore, with less than 60 years left on the lease, the restrictions for purchase are increased and buyers may face limitations on financing (DollarBack Mortgage, 2024). Hence, the property of the value may decline exponentially from then on due to the decrease in value and demand. HDB flat prices have steadily increased over the years, with flats going up to S\$1.7 million in prime estates (Lee J., 2024), which is absolutely mind boggling. Given that HDB flats were made with providing affordable housing for the masses in mind, many complaints have risen over the years about the steadily increasing prices of HDB flats, making citizens question if they are even affordable anymore. Thankfully, some Bayesian statistics may be able to answer some of our doubts.

3. Bayesian Analysis

The main difference between Bayesian methods and its frequentist counterparts is the incorporation of prior knowledge of the model parameters. Model parameters are treated as random variables with their own probability distributions, also known as the prior. These prior distributions can improve the efficiency and accuracy of the estimates, especially when data is

limited. The incorporation of a prior distribution can be useful in situations when there are previous research findings that can support the analysis and when the data simply isn't substantial enough. However, non-informative or weakly informative priors, like a uniform prior, are also often used to allow the data to speak for itself, especially when little to no prior research is available.

The main goal of Bayesian analysis is commonly to find the coveted posterior distribution, which is proportional to the likelihood multiplied by the prior:

$$p(\theta|Y) \propto p(Y|\theta)p(\theta)$$

The posterior distribution provides a comprehensive picture that captures uncertainty in the estimates and can be used to calculate various summary statistics, like the mean and credible interval, which is unlike classical regression where only point estimates and confidence intervals can be derived. Hence, Bayesian statistics often paints a broader and more precise picture about parameters than frequentist statistics. Many times in practical use, however, a closed-form solution of the posterior distribution is unattainable, hence, sampling techniques would need to be used in order to derive the distribution of the estimators.

3.1 Sampling

Common sampling techniques like the Markov Chain Monte Carlo (MCMC) techniques include Gibbs sampler, Metropolis algorithm and Hamiltonian Monte Carlo (HMC). While there is no one-size-fits-all sampler, HMC is often dubbed as the most efficient. Inefficiencies from Gibbs sampler and Metropolis algorithm tend to arise from the random walk behaviour, which can be a problem when dealing with high dimensional data. HMC employs an idea from physics, Hamiltonian dynamics, to restrain the random walk behaviour and allow for faster sampling (Gelman et al., 2021). It enhances the exploration of the target distribution by introducing auxiliary momentum variables, denoted as ϕ for each parameter in the model, denoted as θ . This creates a joint distribution $p(\theta, \phi|y) = p(\phi)p(\theta|y)$, where $p(\phi)$ represents an independent distribution for the momenta. The simulations of interest remain focused on θ , with ϕ serving as an auxiliary variable to facilitate faster movement through the parameter space. HMC leverages on the Hamiltonian dynamics to guide the exploration of the parameter space as the algorithm simulates the movement of a particle in a potential energy landscape, where the particle's position corresponds to the model parameters, θ , and the particle's momentum, ϕ , represents its

velocity. The movement of the particle is simulated using the ‘leapfrog’ integration technique, which breaks down the Hamiltonian equations of motion, allowing for the approximation of the particle’s trajectory. Similar to other algorithms, HMC incorporates an acceptance-rejection step which ensures that the generated samples accurately represent the target posterior distribution by accepting or rejecting proposed moves based on a calculated probability. By computing the gradients of the log posterior density with respect to each parameter, these gradients guide the algorithm in determining the direction and magnitude of the particle’s movement in the parameter space, which is unlike simpler MCMC methods like Gibbs sampling. HMC is an essential tool for addressing the challenges that often arise from complex Bayesian models, especially when handling high-dimensional parameter spaces as its use of momentum variables and Hamiltonian dynamics allows for more efficient exploration of the posterior distribution as compared to traditional MCMC methods. However, on the other hand, HMC can be more computationally expensive and careful tuning is typically needed to optimise performance. But thanks to technology, HMC sampling can be easily incorporated into Bayesian models, even without fully understanding the algorithm behind it.

3.2 Regularisation

Bayesian methods can also easily incorporate regularisation techniques to prevent overfitting, which is a common cause of concern in regression analysis, especially when dealing with a large number of predictors. There are a few factors affecting Bayesian regularisation: the location, scale and form of the prior distribution can affect the degree of regularisation, and how the posterior inference is summarised, e.g. the posterior mode is more regularised than the full posterior. For example, lasso, a common non-Bayesian form of regularisation, employs a prior distribution that is sharply peaked at 0 and estimates coefficients based on its posterior mode (Gelman et al., 2021).

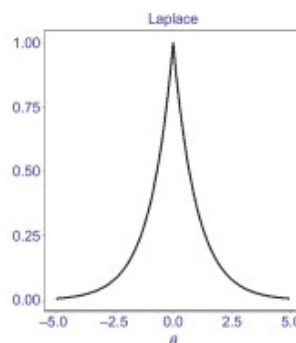


Figure 3.2: Laplace Distribution

The sharp peak at 0 combined with the focus on the posterior mode forces some coefficients to shrink to 0 in settings where the data is very noisy.

Bayesian methods are also good at handling missing data by integrating over the missing values, and hence, are very useful when dealing with sparse data.

3.3 Regression Analysis

There are many well-known Bayesian methods and regression analysis is a very well-travelled topic in statistics, having applications in numerous fields including real estate and geology. It is commonly used to study the relation among 2 or more variables and in general, Bayesian regression is often concerned with the conditional distribution of an independent variable y , given dependent variable x where each observation $(x, y)_i$ is exchangeable.

One such type of regression is logistic regression, where the outcome variable y_i is a binary variable, i.e. y_i can only take on values 0 and 1. Logistic regression has been used throughout many sectors to model various results. For example, it was used to predict breast cancer tumours in the medical field (Lukman, P. A. et al., 2021), and to predict the probabilistic forecasts of the minimum September arctic sea ice cover in the environmental field (Horvath, S. et al., 2020).

The ultimate goal of logistic regression is to model the relationship between predictor variables and the probability of the binary outcome variable, understanding how the parameters shape the probability of the outcome. It uses a sigmoid function, like the logit function, to link the linear combination of the predictor variables to the probability of the outcome, ensuring that the predicted probabilities lie between 0 and 1.

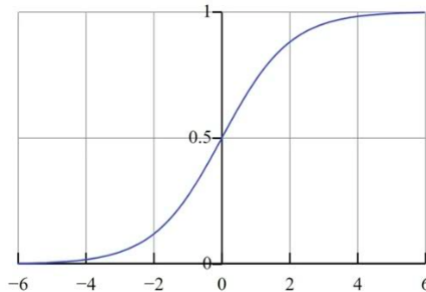


Figure 3.3: Sigmoid function

$$\text{logit}(\theta_i) = \log\left(\frac{\theta_i}{1 - \theta_i}\right) = \beta_0 + \beta X$$

The model parameters, modelled as β above, determine the weight of the relationship between the predictors and the outcome, and are typically estimated using the maximum likelihood estimation, which finds the value of each parameter that maximises the probability of observing the data. Since parameters are treated as random variables in the Bayesian framework, the goal is to also estimate the posterior distribution of each parameter. Now, let's examine how well regression analysis has worked in real estate analysis.

3.4 Bayesian methods in Hedonic Price Modelling

Traditionally, the hedonic price model is used to estimate the value of a property, by considering the extent to which each factor affects the property price. Hedonic price modelling often involves regression analysis to predict prices and relies on the economic theory of decomposing prices into measurable attributes (Hargrave, M. 2020). Conventional approaches to hedonic price modelling include linear regression using ordinary least squares (OLS), that focuses on finding point estimates that minimise the sum of the squared errors between the actual values and the predicted values of the dependent variable, also known as the residuals. With the broad application of hedonic price models among many real estate research, many hedonic price models have arisen over recent years, even those that include Bayesian methods.

It is no surprise that with additional information about the model parameters and more robust model specifications, Bayesian models tend to perform better than their frequentist counterparts. In a study conducted to investigate the use of Bayesian methods for hedonic pricing in Toronto, Canada, it was found that Bayesian spatially varying coefficient process (SVCP) was much better at predicting and estimating housing prices as compared to traditional linear regression models (Wheeler, D. C. et al., 2012). Not only did the Bayesian method outperform in terms of predictive performance, the study also found that the Bayesian approach provided complete inference on all model parameters which offered valuable insights beyond just simple point estimates. This ultimately drives at the interpretation of the results of the analysis -- the posterior distribution of each coefficient for each parameter would provide a range of plausible values for each parameter along with their respective probabilities, allowing for a more nuanced understanding of the relationship between the dependent variables and the independent variables. In another research comparing Bayesian methods to OLS in the presence of spatial autocorrelation in housing data suggests that Bayesian approaches are more effective in

correcting the inefficient OLS estimates brought about by spatial dependencies (Zhou, Q. 2007). This is particularly important in the real estate market analysis where spatial effects are non-trivial and can skew the results of traditional regression models.

Another study revealed that by allowing for varying variances across observations, Bayesian methods are better at handling heteroscedasticity, a common problem in real estate analysis, since such models can account for the non-constant variability in the data, leading to more accurate and reliable estimates (Sun, H. 2003). Heteroscedasticity may render OLS estimates inefficient as it violates the assumption that OLS is the Best Linear Unbiased Estimator (BLUE) where the variance is independent and identically distributed. Since Bayesian methods can directly estimate the heteroscedastic terms from the data without the need to explicitly specify the process of doing so, they are particularly useful when handling complex relationships between location and variance, like in the real estate market.

A study incorporating hierarchical Bayesian methods in the valuation of the Hong Kong housing market has also shown major improvements in comparison to conventional valuations methods like the average per square foot method and a bank's expert assessment, with a consistently lower mean average percent error (MAPE) than the latter 2 (Cheung et al., 2010).

The general consensus is that Bayesian methods offer a powerful and flexible framework for addressing the complexities in real estate data. They can improve the inefficiencies brought about by conventional hedonic price models and provide more precise results which are paramount when making informed decisions in the real estate market.

4. Data and Methods

4.1 Data

There are 2 types of HDB flats that can be purchased in Singapore -- build-to-order (BTO) flats and resale flats, where the former is completely new and the latter is not. It is to be noted that flats can only be sold, with some special exceptions, after meeting the minimum occupation period (MOP), which is typically 5 years, in other words, the remaining lease of resale flats would typically be less than 94 years. Due to limited data however, only resale flat prices will be used for this analysis.

The dataset used in this report shows the transactions of resale flats, taken from the data.gov.sg, from January 2017 to October 2024. The chosen dataset includes the following variables:

month	town	flat_type	block	street_name	storey_range	floor_area_sqm	flat_model	lease_commence_date	remaining_lease	resale_price
2017-01	ANG MO KIO	2 ROOM	406	ANG MO KIO AVE 10	10 TO 12	44	Improved	1979	61 years 04 months	232000
2017-01	ANG MO KIO	3 ROOM	108	ANG MO KIO AVE 4	01 TO 03	67	New Generation	1978	60 years 07 months	250000
2017-01	ANG MO KIO	3 ROOM	602	ANG MO KIO AVE 5	01 TO 03	67	New Generation	1980	62 years 05 months	262000

Figure 4.1.1: Snapshot of the Data

Where each of the variables are represented as follows:

Month: year and month of transaction (date, string)

Town, block and street name: overall location of transacted flat (categorical, string)

Flat type and floor area: size of transacted flat (categorical and numerical, string and integer)

Storey range: storey range of transacted flat (numerical range, string)

Flat model: overall layout of transacted flat (categorical, string)

Lease commence date: year in which the transacted flat was built in (year, string)

Remaining lease: years and months left of the transacted flat (numerical, string)

Resale price: transacted price (numerical, integer)

	floor_area_sqm	lease_commence_date	remaining_lease	resale_price		year	town	26
count	192149.000000	192149.000000	192149.000000	1.921490e+05	count	192149.000000	block	2716
mean	97.044653	1996.140334	74.06100	5.048896e+05	mean	2020.650271	street_name	570
std	24.038112	14.116789	14.03203	1.761366e+05	std	2.220595	flat_type	7
min	31.000000	1966.000000	41.00000	1.400000e+05	min	2017.000000	flat_model	21
25%	82.000000	1985.000000	63.00000	3.750000e+05	25%	2019.000000		
50%	93.000000	1996.000000	74.00000	4.750000e+05	50%	2021.000000		
75%	112.000000	2011.000000	88.00000	6.010000e+05	75%	2023.000000		
max	366.700000	2020.000000	97.00000	1.588000e+06	max	2024.000000		

Figure 4.1.2: General Information about the Data

Before cleaning up the data, we shall first examine the general trends.

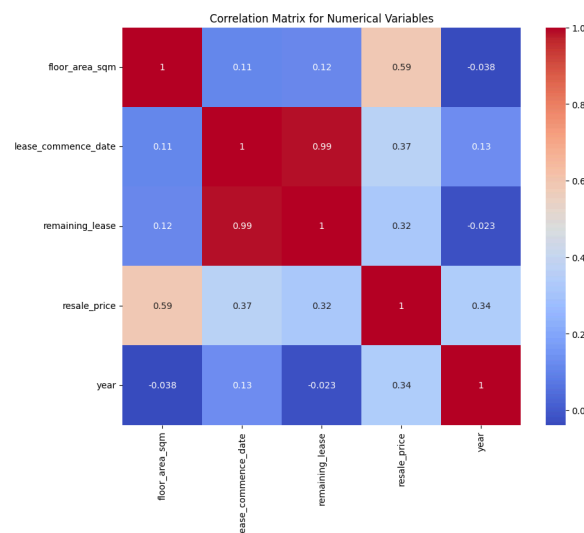
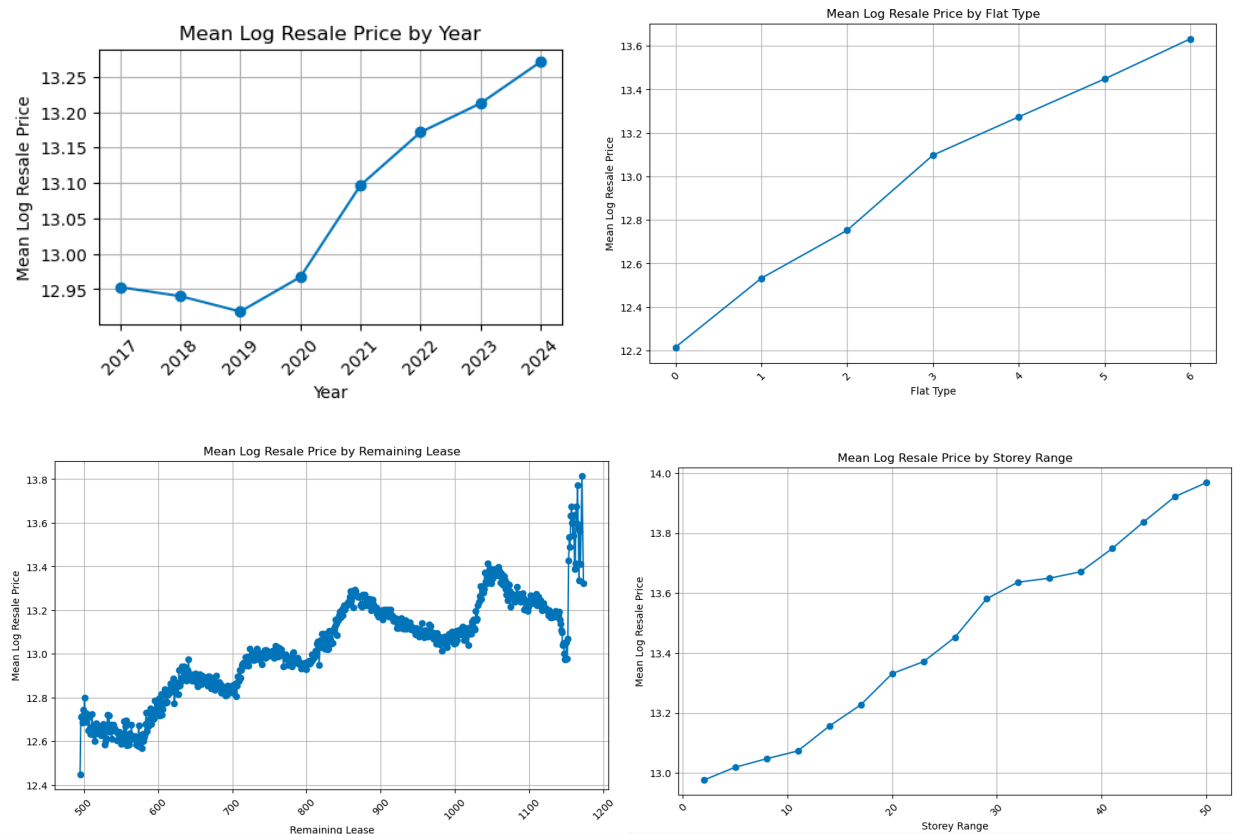


Figure 4.1.3: Correlation Matrix

The correlation matrix for the numerical variables gives us a general idea of where to look for predictor variables.



Figures 4.1.4(a-d) Graphs of various Predictor Variables plotted against Mean log Resale Price

Unsurprisingly, the price of the flat does have a positive relationship with time, flat type, remaining lease and storey range. From the correlation matrix, we can see that there are some independent variables that are highly correlated with one another, like the remaining lease variable and the lease commencement variable, hence a few variables should be dropped.

4.2 Data cleaning

A **year** variable will be made in place of the **month** variable that just captures the year of the transaction, under the assumption that the month-to-month variance is negligible, since Singapore's housing market is not as volatile.

The **block** and **street name** variables will also be dropped and the 26 towns will be summarised into each of their respective **regions**, encoded from 1-6, instead. The regions are ordered in accordance to my own preference, i.e. 6 representing the region I would want to live in the most, and 1 being the region I find least desirable.

The **floor area** of a flat is solely dependent of its flat type and flat model, hence, the floor area variable will be removed, and the **flat type** variable will be converted to a categorical variable that takes on values from 1 to 7, representing flats of 1 to 5 rooms along with executive and multi-generation flats, ordered from smallest to biggest. The **storey range** variable will also be simplified to a numerical storey average variable instead. The **flat model** variable will also be dropped under the assumption that HDB has homogenised flat models for most recent flats, varying only the community care flats which are built specifically for the pioneer generation. HDB has stopped building many sought after flat models, like the Maisonette, Terrace and Design, Build and Sell Scheme (DBSS) flats, and have opted for the more conventional Model A flats instead (PropertyGuru, 2020).

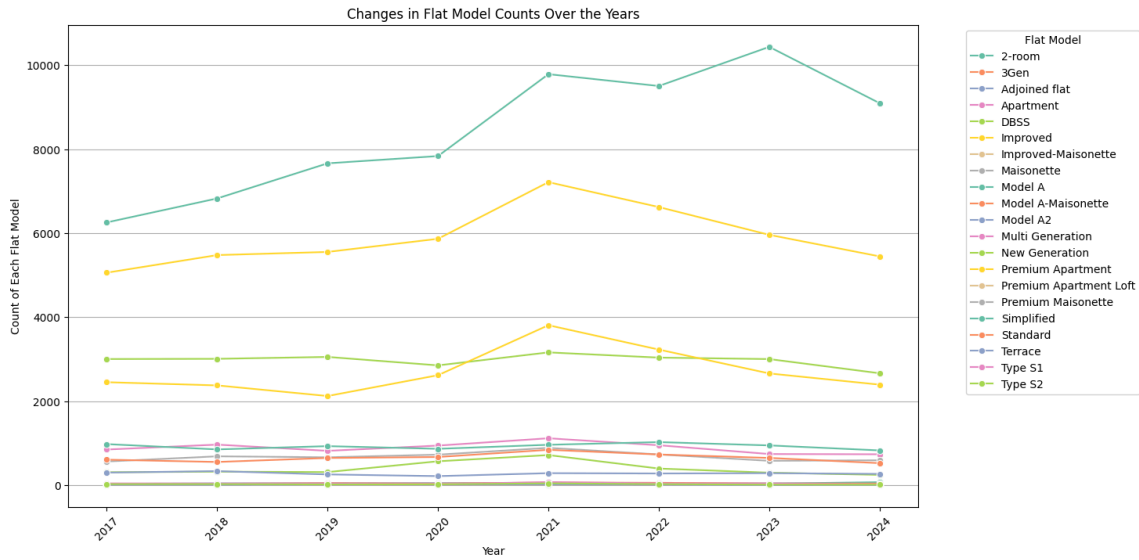


Figure 4.2.1: Changes in Flat Models over the Years

Since the remaining lease is just calculated by subtracting the year in which the flat was sold by the lease commencement date, the **lease commence date** variable will be dropped. The **remaining lease** variable will also be simplified to just the years left instead of years and months. An additional binary variable, **affordability**, will be added to the model. The probability of me being able to afford a flat within my budget would simply be the probability that the price of the flat is less than or equals to \$600,000, an arbitrary budget, i.e.

$$affordability = \begin{cases} 1 & \text{price} \leq 600,000, \\ 0 & \text{otherwise.} \end{cases}$$

Hence, the final dataset we have is as follows:

	flat_type	storey_range	remaining_lease	resale_price	year	affordability	region
0	1	11.0	61	232000.0	2017	1	6
1	2	2.0	60	250000.0	2017	1	6
2	2	2.0	62	262000.0	2017	1	6

Figure 4.2.2: Final Working Data

Before jumping into the model proper, let's review the data again.

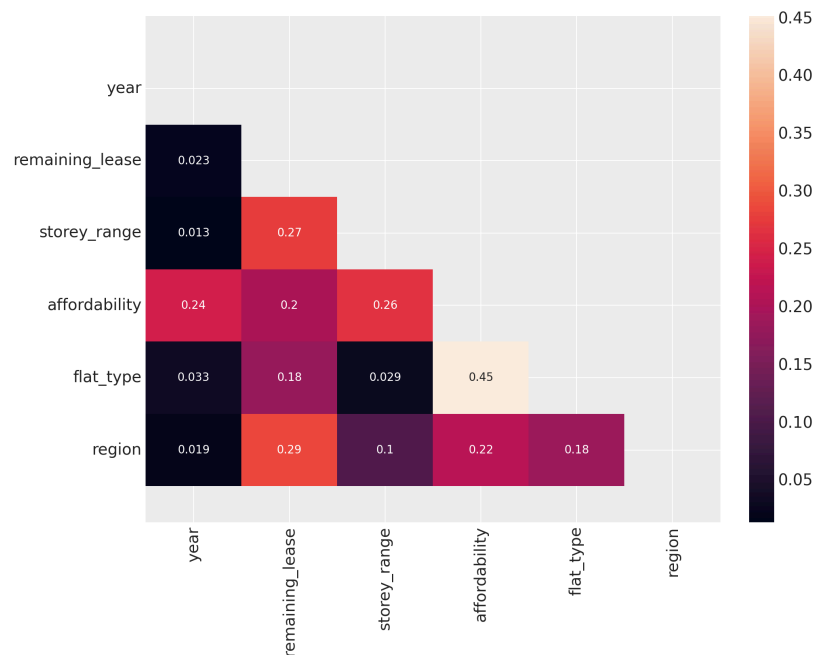


Figure 4.2.3: Correlation Matrix

We can see that the predictor variable that affects affordability most would be the flat type variable, which is in line with the original correlation matrix, since floor area is directly dependent on the type of flat. We should also note that while the correlation is not very high, the remaining lease variable is slightly correlated with all the other predictor variables, which might prove to be troublesome in regression models.

The dataset will be split into training data and test data, at a ratio of 4:1, for cross-validation later on.

4.3 Logistic Regression Model Set-up

A simple model will be used which assumes that the price of a flat would be a function of some flat characteristics, namely region, flat type, storey and remaining lease. To account for the inflation index, the year variable will be included as well.

The logistic regression model can then be defined by the binary response variable y_i , represented by affordability, and the predictor variables $X_1, X_2, \dots, X_5$, represented by region, flat type, storey, year and remaining lease:

$$\text{logit}(\theta_i) = \log\left(\frac{\theta_i}{1 - \theta_i}\right) = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + \beta_5 X_{5i}$$

where $\theta_i = P(y_i = 1)$ and $\beta_0, \beta_1, \dots, \beta_5$ are our unknown parameters.

y_i follows a Bernoulli distribution with $p = \theta_i$ so we have the following likelihood function:

$$p(y_i|\theta) = \prod_{i=1}^n \theta^{y_i} (1 - \theta)^{(1-y_i)}$$

4.4 Priors

Given the large sample size and relatively few number of parameters, weakly-informative priors will be used for all model parameters. Therefore, the prior distribution for β_i is assumed to be normally distributed with mean $\mu_0 = 0$ and a large variance σ_i

$$p(\beta_0, \beta_1, \dots, \beta_5) = \prod_{i=1}^5 \frac{1}{\sigma_i \sqrt{(2\pi)}} \exp\left(-\frac{(\beta_i - \mu_0)^2}{2\sigma_i^2}\right)$$

4.5 Computation

With our likelihood and priors, we can obtain the posterior distribution for our model parameters. When the posterior distribution is attainable in its closed form, it can be represented by the probability density function of a specific distribution and conclusions can be drawn more easily, without complicated sampling techniques. However, in this case, with a non-conjugate, weakly informative prior, the analytical form is hard to attain. Hence, sampling methods will be used to sample from the posterior distributions of the parameters instead. The default sampling method for Python's probabilistic programming is NUTS (No U-Turn Sampler) which is a form of HMC sampling. Libraries like PyMC provide a convenient way to sample from the models and analyse the resulting posterior distributions.

5. Results from Logistic Regression

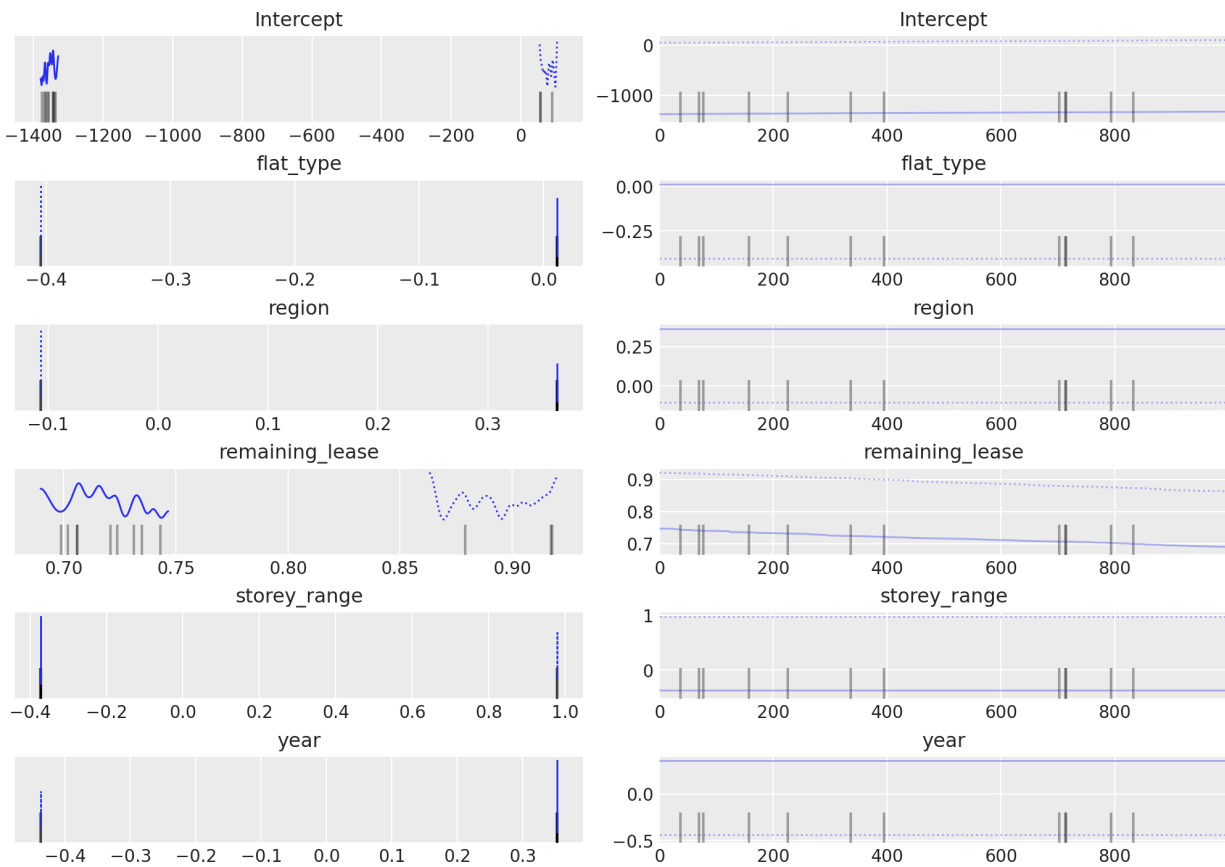


Figure 5.1: Posterior Distributions of each Parameter in the 1st Run

It is obvious that the model did not converge, resulting in a questionable set of “distributions” for each parameter. Upon further inspection, it seems like the correlation between the remaining lease variable and the other predictors might be affecting the convergence of the model.

5.1 The Issue with Highly Correlated Predictors

Highly correlated parameters pose significant challenges in statistical models, especially in Bayesian models, where accurate and efficient exploration of the posterior distribution is important, as they slow down the effectiveness of sampling algorithms in navigating the posterior parameter space. Additionally, these parameters reduce the effective sample size of the model due to the non-independence of each consecutive sampling. Apart from slow convergence, or not converging at all, simulation inference from correlated draws are also often not as precise as the inference drawn from the same number of independent draws. These parameters also make

it difficult to isolate the individual effects of each parameter on the independent variable and the inferences drawn from the model are highly sensitive to the model's assumptions, which can lead to large standard errors for parameter estimates, increasing uncertainty. Not to mention, parameters that are highly correlated require more sophisticated and computationally intensive sampling algorithms in order for sufficient exploration of the posterior distribution, which increases the computational burden and time required for the analysis.

The solution to high correlated predictors comes in the form of reparameterization or the use of more advanced sampling algorithms (Gelman et al., 2021). However, due to computational constraints, advanced algorithms will not be explored in this report.

5.2 Trying Again

Just to test out if the remaining lease variable was the main culprit behind the non-convergence, the model was run without this variable. And indeed, after removing the remaining lease variable, the model reached convergence.

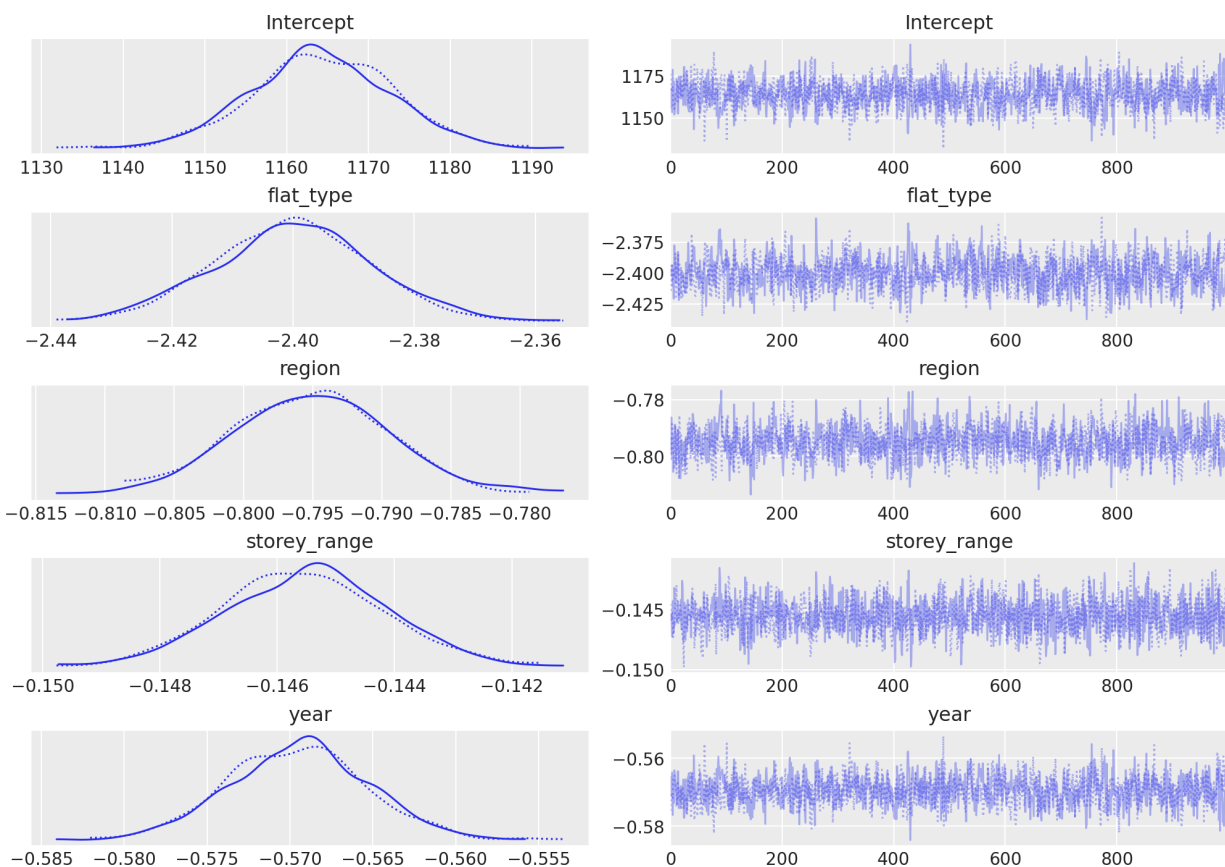


Figure 5.2.1: Posterior Distribution of each Parameter in the 2nd Run

Now, since 1 of the 2 ways proposed is not feasible, we are only left with 1 choice -- reparameterization. In the context of HMC, transformations are often employed to address the issue of parameters having very short-tailed or long-tailed distributions, which can negatively impact convergence (Gelman et al., 2021). From the original working data set in Figure 2.2, we can see that the range of values for each variable differs by a large margin. There are different ways of reparameterization, but perhaps the most straightforward way would be to standardise the data by centering it at the mean and scaling it to unit variance:

$$\text{standardised value} = \frac{\text{value} - \text{mean}}{\text{standard deviation}}$$

After reparameterizing all the numerical variables, i.e. remaining lease, year, and storey range, the model was run again, with 4 independent sequences now.

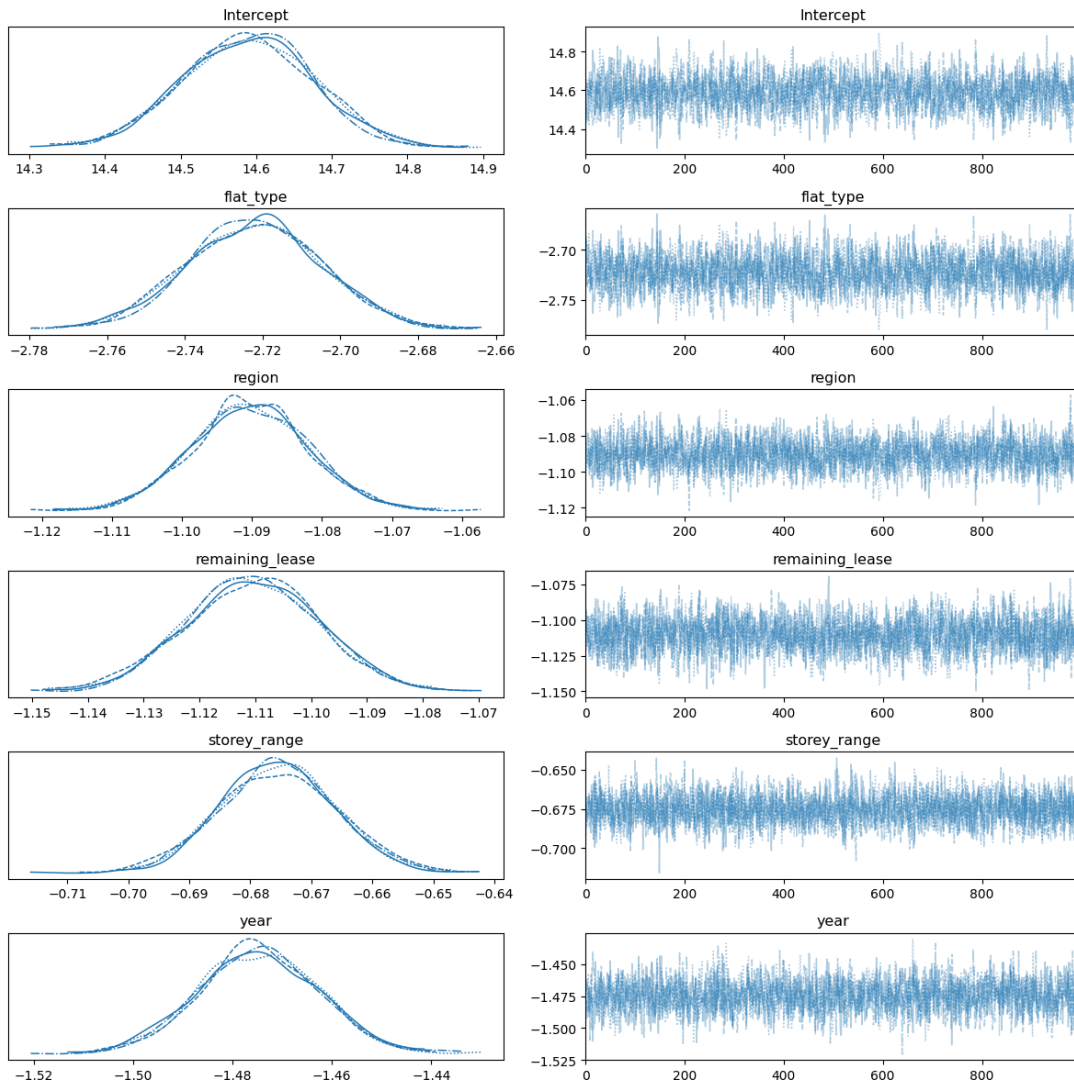


Figure 5.2.3: Posterior Distribution of each Parameter in the 3rd Run

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
Intercept	14.589	0.086	14.429	14.748	0.002	0.001	1840.0	2327.0	1.0
flat_type	-2.722	0.017	-2.753	-2.689	0.000	0.000	2140.0	2839.0	1.0
region	-1.090	0.008	-1.105	-1.074	0.000	0.000	2132.0	2492.0	1.0
remaining_lease	-1.111	0.012	-1.132	-1.088	0.000	0.000	2306.0	2590.0	1.0
storey_range	-0.675	0.009	-0.693	-0.658	0.000	0.000	3118.0	2829.0	1.0
year	-1.475	0.012	-1.498	-1.453	0.000	0.000	2326.0	2749.0	1.0

Figure 5.2.4: Summary statistics for Posterior Distribution

All values of r-hat equal to 1 which suggests that the algorithm has converged and we have the posterior distribution for our parameters. All variables have negative coefficients which signifies a negative relationship with affordability, which is in line with my initial beliefs. From the summary statistics, we can see just how much each predictor variable affects flat prices, with flat type having the most impact and storey range with the least.

Let's say I have a goal of buying a 4-room resale flat on the 20th floor with a remaining lease of 90 years, in region 6, in 2030. The model estimates the probability of affordability to be around 0.0125%, which is depressing to say the least, but not all that surprising.

6. Model Checking and Evaluation

Now, we need to check if our model is robust enough, and look for any room for improvement. Model checking provides us an overall look of how our model is doing -- it is good to check what aspects are not being captured by the model since it is difficult to include every single thing in the first place. The goal of model checking is often not to compare and pick an ideal model, but to evaluate the consistency of the model with observed data, understand what it is lacking and potentially improve on it. The ideal model check would be done by setting up a comprehensive joint distribution of all possible models that fit the data, however, that is not computationally feasible. There are a few different ways to carry out model checking, some of which include external validation and posterior predictive checks.

External validation involves the model making predictions about future data, and then comparing it to actual observations. However, it is not always feasible to collect more data -- consider a study on suicide rates. There are several ways of approximating the predictive accuracy of a model even without more data. A few of these approximation methods include the within-sample predictive accuracy, adjusted within-sample predictive accuracy and cross-validation.

Within-sample predictive accuracy is the most straightforward approach, using the log pointwise predictive density (lppd) calculated on the observed data. While this method is simple and easy to understand, it runs into the problem of overestimating the true predictive accuracy since it was evaluated on the data from which the model was built on. Adjusted within-sample predictive accuracy is similar to the within-sample predictive accuracy approach, but with an added penalty term to correct the biases of the lppd. Lastly, cross-validation aims to separate the data into a training and holdout set, mimicking the process of predicting new data since the model has not been exposed to the holdout set during training.

Additionally, posterior predictive checks offer an alternative approach to model checking.

Posterior predictive checks are a fundamental aspect of Bayesian modelling as they help assess the model by comparing observed data and data simulated from the model's posterior predictive distribution. Posterior predictive checks use test quantities to quantify the differences between observed data and the model's predictions -- these test quantities can be numerical summaries, or even graphical displays to reveal patterns between the observed and simulated data. The choice of test quantities should be guided by the specific aspects of the data that are of interest and the potential shortcomings of the model. The differences between the observed test quantity and its posterior predictive distribution can also be quantified by a p-value. The p-value represents the probability of observing a test quantity as extreme as, or more extreme than the one calculated from the actual data, assuming that the model is correct and a p-value near 0 or 1 indicates a lack of fit. P-values however, should be interpreted as probabilities rather than as a strict decision threshold.

6.1 Holdout Valuation

Posterior predictive checks are made simple with many probabilistic programming libraries like the aforementioned PyMC, however, it is quite computationally expensive and might not be for your (my) average computer. As such, holdout evaluation will be used to evaluate this model.

From Figure 4.1, we can see that the actual and predicted values seem pretty similar. The model achieved a log loss of 0.265, which indicates a relatively good prediction accuracy. However, due to the lack of information, there isn't exactly an established benchmark for what a good log loss score is in this specific application.

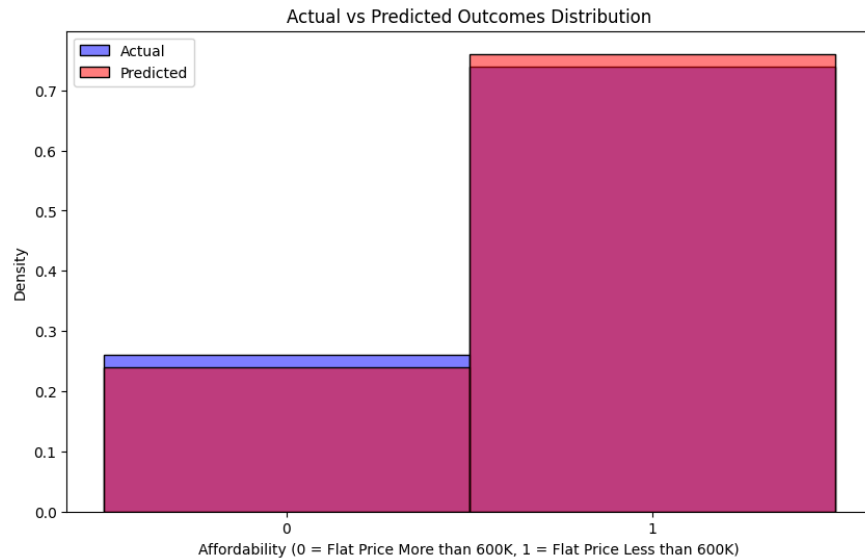


Figure 6.1: Graph of Densities between Actual and Predicted Outcomes

7. Discussion and Conclusion

A very simple logistic regression model was employed to model the probability that I will be able to afford a house in the future, under some aforementioned assumptions. While the model ran into some issues concerning the correlation between predictor variables, reparameterization had resolved the problem. The results have shown that while all the predictor variables are negatively correlated with price, flat type has the biggest impact on the price of a resale HDB flat, and storey has the least impact. The holdout valuation has also shown that the model achieved a log loss score of 0.265, which is a seemingly “good” score.

7.1 Discussions and Limitations

Many assumptions were made in the building of this model. For example, the flat model variable was dropped under the assumption that it will not matter for years to come due to the homogeneity of recent HDB flats. However, a prior study has shown that the flat model does have an impact on price (Lee, M. 2021). Although that might be due to the relatively high price tag that the discontinued models, like the Terrace model, take on, keeping this variable in may result in a more comprehensive model.

Additionally, for a more robust model, factors like income, amount of savings and marital status could have also been included such that an arbitrary budget does not need to be set, i.e. it will just be the probability that the flat price, as a function of the characteristics, will be less than budget, which can be some function of income and savings. However, the acquisition of income and savings data might prove to be difficult.

One other major simplification of this model comes from the grouping of each town into a specific region. The location of the flat could have been represented by longitudes and latitudes instead, which can better account for the proximity between important facilities like MRT stations and shopping malls. In doing so, other factors, like the distance to an MRT station, can be modelled and is no doubt, a major factor in determining price.

The towns that each flat resides in need not be grouped into regions as well, and can be modelled in their own cluster, using hierarchical modelling. The prices of flats within the same town are likely more similarly distributed than the prices of flats within another town, violating the independence assumption. The first level of the hierarchical model would encompass all the flat characteristics, like the type, storey and remaining lease, while the second level would encompass town specific characteristics like the amenities present and the nearest MRT station. By estimating the variance of the town specific coefficients, the model can determine the extent to which the flat characteristics vary across towns. The model may look something like the following:

$$\text{logit}(\theta_{ij}) = \log\left(\frac{\theta_{ij}}{1 - \theta_{ij}}\right) = \beta_{0j} + \beta_{1j}X_{1i} + \beta_{2j}X_{2i} + \beta_{3j}X_{3i} + \beta_{4j}X_{4i} + \beta_{5j}X_{5i}$$

where $\beta_{ij} = \gamma_{i0} + \mu_{ij}$ and γ, μ are hyperparameters

Interaction terms between predictor variables can also be added to supplement the model. A unit change in flat type may depend on the value of which the remaining lease is fixed at. If that is the case, then an interaction term should be included in the model by adding the cross-product term $(x_i - \bar{x}_i)(x_j - \bar{x}_j)$ as an additional predictor and treat these interactions like any other predictor variable.

The model could have also included some non-linear transformation of certain variables if the relation between the independent and dependent variable is not as clearcut. For example, many suspect that HDB flat prices may be increasing more than proportionally with time, hence,

adding a non-linear term, such as a quadratic term, of the year variable may improve the fit of the model.

7.2 Conclusion

To conclude, I believe that while this model is not the most robust and has its aforementioned limitations, it is a good stepping stone for future analysis on the HDB market. By knowing how much to set aside for one of the first few big ticket items of one's life and how various factors affect flat prices, more informed financial decisions can be made to avoid financial distress.

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